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Project Titles and Descriptions

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Kiray: Local Asset Rental and Sharing Platform

1. Introduction

Kiray connects asset owners with borrowers to rent or share valuable items within local communities or businesses. The platform focuses on high-demand, high-cost, or short-term use items, reducing ownership costs while maximizing utilization. Common example asset categories include **construction materials and machinery, clothing and furniture for events like weddings and funerals**, **audio equipment and speakers and musical instruments.**

The system has two main components: a web dashboard for managing listings, bookings, and payments, and a mobile-friendly interface for browsing and requesting assets. AI features enhance personalization, automation, and trust, creating a modern, intelligent rental ecosystem.

2. Problem Statement

In Ethiopia, individuals and small businesses face high costs for assets used occasionally, such as machinery, audio equipment, or formal/traditional event attire. Current rental options are informal, fragmented, and unreliable. Key challenges include:

- High upfront cost for items needed occasionally
- Difficulty finding available items locally
- Limited trust in asset condition and owners

Kiray addresses these issues by combining **structured listings, secure booking, and AI-powered features** to make rentals **efficient, reliable, and adaptive**.

3. System Overview and Features

A. Web Dashboard

- **Profile Management:** Users create verified profiles with location, preferences, and payment details.
- **Asset Management:** Owners upload assets with photos, category, condition, availability, and rental price. Categories include construction tools, furniture, audio systems, musical instruments, and traditional wedding attire. Borrowers can search and filter listings.
- **Booking & Payments:** Rental requests, approval workflows, deposits, and digital payments are supported.

B. Mobile Interface

- Quick browsing, booking, and notifications for new or approved rentals.

C. AI Integration

- **Personalized Recommendations:** Suggests assets based on past rentals, location, and category preferences.
- **Image-Based Tagging:** Automatically categorizes assets and assesses condition from photos.
- **Pricing Suggestions:** Recommends competitive rental prices using historical data and local demand.
- **Trust & Fraud Detection:** Flags suspicious listings or risky behavior to enhance reliability.

4. Conclusion

Kiray creates a trusted, AI-enhanced rental ecosystem in Ethiopia, enabling local communities to efficiently share and monetize assets. By focusing on construction equipment, event materials, musical instruments, and formal/traditional attire, combined with web, mobile, and AI features, the platform delivers smart, safe, and personalized rentals while creating new economic opportunities for owners.

Med-Trace (A Real-Time Pharmacy Inventory Synchronization and Medicine Discovery Platform)

Med-Trace is a pharmaceutical discovery and inventory synchronization platform designed to bridge the information gap between patients and pharmacies. The system consists of a cross-platform mobile application built with Flutter for patients and pharmacy staff, and a robust administrative dashboard powered by Laravel Filament. The platform facilitates a real-time workflow where patients search for scarce medications, verify availability via geolocation, and reserve items using Telebirr/CBE Birr. By digitizing the medicine search process, Med-Trace ensures that life-saving drugs are found efficiently while providing pharmacies with a digital sales channel.

2. Problem Statement

In Ethiopia, the pharmaceutical market is highly fragmented, and medication shortages are common. Patients currently rely on "physical searching" manually visiting multiple pharmacies to find specific drugs which leads to critical delays in treatment, high transportation costs, and emotional distress. Conversely, pharmacies lack a way to broadcast their current stock to the public, leading to inventory inefficiencies. Existing methods lack real-time accuracy and price transparency. Med-Trace addresses these gaps by providing a centralized, verified inventory database that saves time and potentially saves lives.

3. System Overview and Features

A. Mobile Application

The app serves two distinct user roles: Patients (Searchers) and Pharmacy Staff (Providers).

1) User Profiles & Authentication

- **Patient Profile:** Personal info, location settings, and prescription history.
- **Pharmacy Profile:** Location (GPS), license verification status, and contact details.

- **Security:** Phone-based authentication with role-based access control.

2) Medicine Discovery & Search

- **Smart Search:** Search by brand name, generic name, or category.
- **Prescription OCR:** A built-in feature allowing users to upload a photo of a handwritten prescription. The system uses **Optical Character Recognition (OCR)** to extract medicine names and initiate a search automatically.
- **Geo-Filtering:** Results are sorted by proximity, showing the nearest pharmacy with the item in stock.

3) Reservation & Workflow

- **Availability Status:** Real-time stock indicators (In Stock, Limited, Out of Stock).
- **Secure Reservation:** Patients can reserve a medication for a limited window (e.g., 2 hours) to ensure it isn't sold before they arrive.
- **Payment Integration:** Supports a "Commitment Fee" or full payment via **Telebirr/CBE Birr** to secure the reservation.
- **Workflow States:** `Requested` → `Confirmed by Pharmacy` → `Reserved` → `Picked Up/Completed` or `Expired`.

4) Notifications

- **Restock Alerts:** Patients can "Follow" a specific out-of-stock medicine and receive push notifications the moment a nearby pharmacy updates its inventory.
- **Order Updates:** Real-time alerts for reservation approvals or expiration warnings.

B. Admin & Pharmacy Dashboard

Built with **Laravel Filament** for high-speed management of medical data and users.

- **Inventory Management:** Pharmacies can bulk-upload CSVs or manually update stock levels and prices.

- **Request Management:** A dedicated view for pharmacy staff to approve or reject incoming reservation requests.
- **Shortage Analytics:** An admin view for health officials to see "Heatmaps" of which medicines are most searched for but not found (identifying national shortages).
- **System Governance:** Manage verified pharmacy licenses, audit logs, and user reports.

C. Platform & Security

- **Backend:** Laravel REST API with Sanctum for secure token-based communication.
- **Database:** PostgreSQL with **PostGIS** for high-performance geographic coordinate queries.
- **Data Integrity:** Strict validation to ensure only licensed pharmacies can list prescription-only medications.
- **Privacy:** Encryption of patient prescription uploads to ensure medical confidentiality.

4. Conclusion

By replacing the manual, fragmented search for medicine with a streamlined, GPS-enabled digital platform, Med-Trace optimizes the pharmaceutical supply chain in Ethiopia. The platform provides patients with immediate transparency and providers with a wider customer base. With the integration of OCR technology and local mobile payment gateways, Med-Trace offers a technically sophisticated solution that is both commercially viable and socially impactful, ensuring that the right medicine reaches the right patient at the right time.

Local Services Aggregator: On-Demand Local Services Marketplace and Workflow System

1. Introduction

The Local Services Aggregator connects customers with local providers offering services such as plumbing, electrical work, cleaning, and repairs. The platform includes a mobile application built with Flutter for both customers and providers, alongside an admin dashboard powered by Laravel Filament. It supports a complete end-to-end workflow: customers post jobs, providers express interest, selections are made, work is completed, and ratings are given. Real-time in-app notifications and role-based actions ensure transparency, clarity, and trust throughout the process.

2. Problem Statement

Finding reliable local service providers is often fragmented, with inconsistent quality and communication. Customers lack visibility, while providers spend time on irrelevant leads. Existing solutions rarely offer structured workflows or trustworthy rating systems, leaving both sides frustrated. The Local Services Aggregator addresses these gaps by streamlining discovery, ensuring relevant matches, and maintaining accountability through verified ratings.

3. System Overview and Features

A. Mobile App

The app is the primary interface for creating and fulfilling service requests.

1) User Profiles

- Customer: address, additional info.
- Provider: bio, address, selected job types, aggregate rating.
- Phone/password auth; role-based navigation.

2) Jobs

- Customer: create job (title, description, job type, price).
- Auto-match providers by job type; create requested job links; notify providers.
- Customer: list own jobs; view details (type, assigned provider, status, rating flag).
- Provider: requested jobs feed; express interest.
- Customer: view interested providers; select provider.

3) Workflow & Notifications

- States: open → in_progress → provider_done → completed/cancelled.
- Actions: customer selects provider; provider marks done; customer confirms completion; customer can cancel (open/in_progress).
- Notifications: new job (to providers), interest received, selection, status changes, cancellations; mark as read; paginated lists.

4) Ratings

- Post‑completion rating (1–5, optional comment).
- One rating per job; provider's average auto‑updated.

B. Admin Dashboard

Built with Laravel Filament for operational control.

- Manage: users, profiles, job types, jobs, ratings.
- Monitor: notifications, lifecycle distributions.
- Governance: role‑based access; planned audit logs; queue‑ready design.

C. Platform & Security

- Laravel API + Sanctum tokens; strict role checks.
- Core models: User, CustomerProfile, ProviderProfile, JobType, Job, Notification, Rating, pivots for provider–job and provider–jobType.
- Pagination across jobs, job types, notifications.
- Responses hide sensitive IDs; return necessary context.

4. Conclusion

By offering a streamlined, role-aware workflow with reliable notifications and ratings, the platform ensures customers receive efficient services and providers gain relevant job opportunities. The admin dashboard provides full operational oversight, maintaining quality and scalability across the system.